

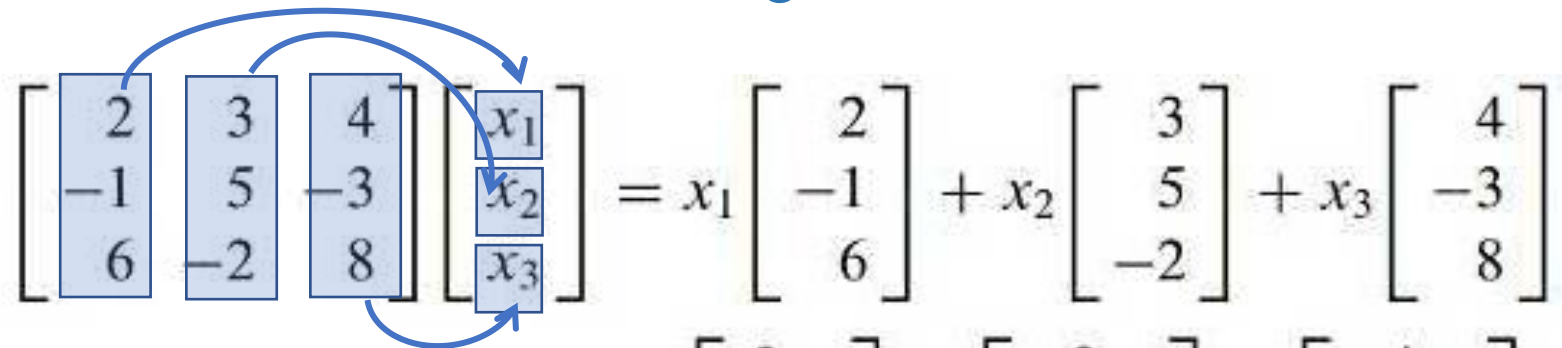
Let A be an $m \times n$ matrix. Then the following statements are logically equivalent. That is, for a particular A , either they are all true statements or they are all false.

- a. For each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
- b. Each $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of the columns of A .
- c. The columns of A span $\mathbb{R}^m$.
- d. A has a pivot position in every row.

Computation of Matrix-Vector Product $A\mathbf{x}$

Compute $A\mathbf{x}$, where $A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

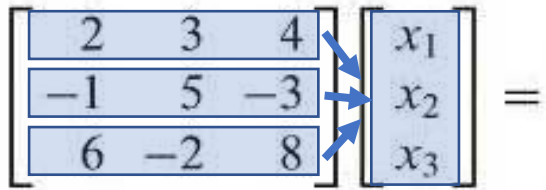
We use $\mathbf{x}$ to generate a linear combination of the columns of A .


$$\begin{aligned} \begin{bmatrix} \boxed{2} & \boxed{3} & \boxed{4} \\ \boxed{-1} & \boxed{5} & \boxed{-3} \\ \boxed{6} & \boxed{-2} & \boxed{8} \end{bmatrix} \begin{bmatrix} \boxed{x_1} \\ \boxed{x_2} \\ \boxed{x_3} \end{bmatrix} &= x_1 \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ -3 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} 2x_1 \\ -x_1 \\ 6x_1 \end{bmatrix} + \begin{bmatrix} 3x_2 \\ 5x_2 \\ -2x_2 \end{bmatrix} + \begin{bmatrix} 4x_3 \\ -3x_3 \\ 8x_3 \end{bmatrix} \\ &= \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{bmatrix} \end{aligned}$$

Computation of Matrix-Vector Product $A\mathbf{x}$

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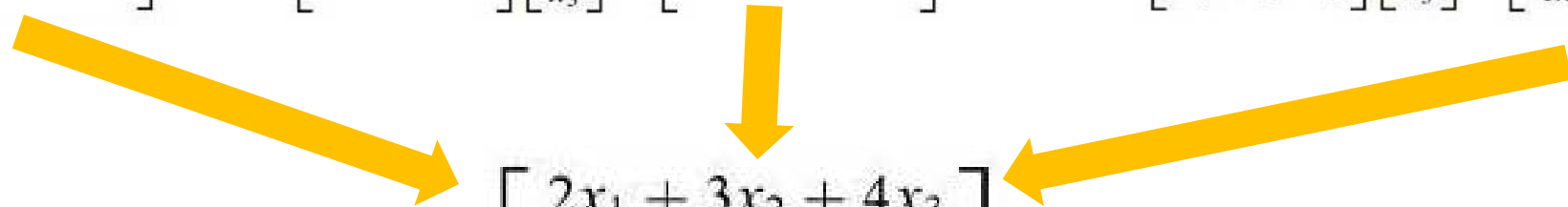
we apply the dot product operation (sum of products) between every row of A and the vector $\mathbf{x}$


$$\begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} =$$

$$\begin{bmatrix} 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 5 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -x_1 + 5x_2 - 3x_3 \end{bmatrix}$$

$$\begin{bmatrix} 6 & -2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6x_1 - 2x_2 + 8x_3 \end{bmatrix}$$


$$\begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -5 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 \cdot 4 + 2 \cdot 3 + (-1) \cdot 7 \\ 0 \cdot 4 + (-5) \cdot 3 + 3 \cdot 7 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -3 \\ 8 & 0 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \cdot 4 + (-3) \cdot 7 \\ 8 \cdot 4 + 0 \cdot 7 \\ (-5) \cdot 4 + 2 \cdot 7 \end{bmatrix} = \begin{bmatrix} -13 \\ 32 \\ -6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r \\ s \\ t \end{bmatrix} = \begin{bmatrix} 1 \cdot r + 0 \cdot s + 0 \cdot t \\ 0 \cdot r + 1 \cdot s + 0 \cdot t \\ 0 \cdot r + 0 \cdot s + 1 \cdot t \end{bmatrix} = \begin{bmatrix} r \\ s \\ t \end{bmatrix}$$



This is a special type of matrices called the **identity matrix** and is denoted by I or I_n (in this example I_3)

$$I_n \mathbf{x} = \mathbf{x} \text{ for every } \mathbf{x} \text{ in } \mathbb{R}^n$$

Properties of the Matrix-Vector Product $A\mathbf{x}$

If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then:

- a. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$;
- b. $A(c\mathbf{u}) = c(A\mathbf{u})$.

Solution Set of Linear Systems

Homogeneous Linear Systems: System of linear equations that can be written in the form

$$A\mathbf{x} = \mathbf{0}$$

where A is an $m \times n$ matrix and $\mathbf{0}$ is the zero vector in $\mathbb{R}^m$.

A homogeneous Linear System *always* has *at least one* solution, which is $\mathbf{x} = \mathbf{0}$ (the zero vector in $\mathbb{R}^m$) and we call it the **trivial solution**

The important question is:

Does a nontrivial solution exist for a homogeneous linear system ?

Do you
Remember
this
theorem ?

Existence and Uniqueness Theorem

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column—that is, if and only if an echelon form of the augmented matrix has *no* row of the form

$$[0 \quad \cdots \quad 0 \quad b] \quad \text{with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

Does a **nontrivial solution** exist for a homogeneous linear system ?

If it is the case (i), then we have only ONE (unique) solution and it must be the **trivial solution** (i.e., **0**) because it is a homogeneous linear system. So the answer is **NO there is no nontrivial solution**

If it is the case (ii), then we have INFINETLY MANY solutions. So, we have the **trivial solution** plus other **nontrivial solutions** So the answer is **YES there is a nontrivial solution**

Does a **nontrivial solution** exist for a homogeneous linear system ?

The homogeneous equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution if and only if the equation has at least one free variable.

Example: Determine if the following homogeneous system has a nontrivial solution. Then describe the solution set.

$$\begin{aligned} 3x_1 + 5x_2 - 4x_3 &= 0 \\ -3x_1 - 2x_2 + 4x_3 &= 0 \\ 6x_1 + x_2 - 8x_3 &= 0 \end{aligned}$$

This is Echelon Form:

C1 & C2 are pivot column $\rightarrow x_1$ & x_2 are basic variables

C3 is not a pivot column $\rightarrow x_3$ is a **free** variable
Since there is at least one free variable, then a nontrivial solution exists.

$A\mathbf{x} = \mathbf{0}$?

$$\begin{bmatrix} 3 & 5 & -4 & 0 \\ -3 & -2 & 4 & 0 \\ 6 & 1 & -8 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & -9 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 3 & 5 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_2 = R_2 + R_1$
 $R_3 = R_3 + 2R_1$

$$\begin{bmatrix} 3 & 0 & -4 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 = R_1 - 5/3 R_2} \begin{bmatrix} 1 & 0 & -4/3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_1 = 1/3 R_1$
 $R_2 = 1/3 R_2$

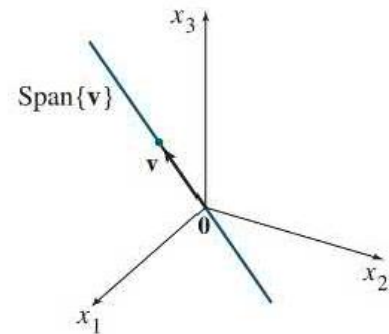
Reduced echelon form

$$\begin{aligned} x_1 - \frac{4}{3}x_3 &= 0 \\ x_2 &= 0 \\ 0 &= 0 \end{aligned}$$

Solution

$$\begin{cases} x_1 = \frac{4}{3}x_3 \\ x_2 = 0 \\ x_3 \text{ is free} \end{cases}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$



Check

$$A\mathbf{x} = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

Example: Describe all solutions of the homogeneous "system"

$A\mathbf{x} = \mathbf{0}$?

$$10x_1 - 3x_2 - 2x_3 = 0$$

$$\begin{bmatrix} 10 & -3 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This is already in Echelon Form:

C1 is a pivot column $\rightarrow x_1$ is a basic variable

C2 & C3 are not a pivot columns $\rightarrow x_2$ & x_3 are free variables

Since there is at least one free variable, then a nontrivial solution exists.

Solution

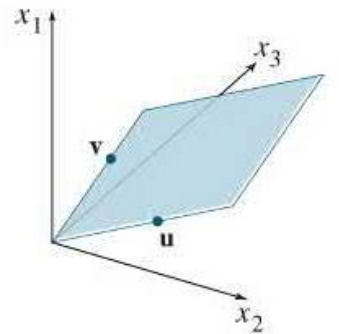
$$\begin{cases} x_1 = .3x_2 + .2x_3 \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{cases}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} .3x_2 + .2x_3 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} .3x_2 \\ x_2 \\ 0 \end{bmatrix} + \begin{bmatrix} .2x_3 \\ 0 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} .3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} .2 \\ 0 \\ 1 \end{bmatrix} \quad (\text{with } x_2, x_3 \text{ free})$$

$\uparrow \quad \uparrow$
 $\mathbf{u} \quad \mathbf{v}$

Parametric vector form

There are infinitely many solutions, and every solution is a linear combination of the vectors $\mathbf{u}$ and $\mathbf{v} \rightarrow$ the solution set is $\text{Span} \{\mathbf{u}, \mathbf{v}\}$.



Check

$$A\mathbf{x} = \begin{bmatrix} 10 & -3 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \left(x_2 \begin{bmatrix} .3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} .2 \\ 0 \\ 1 \end{bmatrix} \right) = x_2 \begin{bmatrix} 10 & -3 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} .3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 10 & -3 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} .2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

Fact

the solution set of a homogeneous equation $A\mathbf{x} = \mathbf{0}$ can always be expressed explicitly as $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ for suitable vectors $\mathbf{v}_1, \dots, \mathbf{v}_p$. If the only solution is the zero vector (**trivial Solution**), then the solution set is $\text{Span}\{\mathbf{0}\}$.

Solution Set of Linear Systems

Nonhomogeneous Linear Systems: System of linear equations that can be written in the form

$$A\mathbf{x} = \mathbf{b}$$

When a nonhomogeneous linear system is consistent, the general solution can be written in **parametric vector form** as:

One vector + **a linear combination of vectors satisfying the corresponding homogeneous system $A\mathbf{x} = \mathbf{0}$.**

EXAMPLE 3 Describe all solutions of $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix} \quad \mathbf{x} = ?$$

$$\begin{bmatrix} 3 & 5 & -4 & 7 \\ -3 & -2 & 4 & -1 \\ 6 & 1 & -8 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -\frac{4}{3} & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{array}{rcl} x_1 & -\frac{4}{3}x_3 & = -1 \\ x_2 & & = 2 \\ 0 & & = 0 \end{array}$$

Solution $\begin{cases} x_1 = -1 + \frac{4}{3}x_3 \\ x_2 = 2 \\ x_3 \text{ is free} \end{cases}$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 + \frac{4}{3}x_3 \\ 2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{4}{3}x_3 \\ 0 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix}$$

$\uparrow \quad \quad \uparrow$
 $\mathbf{p} \quad \quad \mathbf{v}$

$$\mathbf{x} = \mathbf{p} + x_3 \mathbf{v}$$

Check

$$A\mathbf{x} = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \left(\begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \begin{bmatrix} \frac{4}{3} \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \mathbf{b} + \mathbf{0}$$

WRITING A SOLUTION SET (OF A CONSISTENT SYSTEM) IN PARAMETRIC VECTOR FORM

1. Row reduce the augmented matrix to reduced echelon form.
2. Express each basic variable in terms of any free variables appearing in an equation.
3. Write a typical solution $\mathbf{x}$ as a vector whose entries depend on the free variables, if any.
4. Decompose $\mathbf{x}$ into a linear combination of vectors (with numeric entries) using the free variables as parameters.